北京大学
PEKING UNIVERSITY

Introduction to Foundation Models

Kun Yuan

Center for Machine Learning Research @ Peking University

2026 大模型与信息决策课程微信群

Contents

- **Foundation Models**

  Linear regression； Logistic regression; Multi-class regression; Neural network

  Word embedding； Language models; RNN; Seq2seq; Transformer; Decoder-only LLMs; MOE

- **Pre-training**

  Basics in optimization; Gradient descent; Forward-backward propagation; Stochastic gradient descent;

  Adaptive gradient descent; Distributed pretraining; Mixed-precision pre-training; FlashAttention

- **Post-training and Inference**

  SFT; RLHF; PEFT; Alignment; KV Cache

- **Application**

  Chain of thought; Tree of though; Prompt Engineering; Retrieval -agmented generation; **LLM agents**

Grading policy

- **Homework (~30%):**
  Including theoretical assignments and practical coding assignments. For the theoretical assignments, students are required to submit the paper version (either handwritten or printed) in class for attendance purposes. Students who submit their work on time can earn at least 60% of the total score for each individual assignment.

- **Group project (~30%):**
  Including a presentation and a written report. The specific requirements and submission time are yet to be determined.

- **Final Exam (~40%):**
  Including multiple-choice questions, short-answer questions and comprehensive questions, with the main focus on theoretical contents.

Teaching assistants

宋奕龙

Leading TA

吴洲同

李代波

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A Brief Introduction to Deep Learning

Kun Yuan

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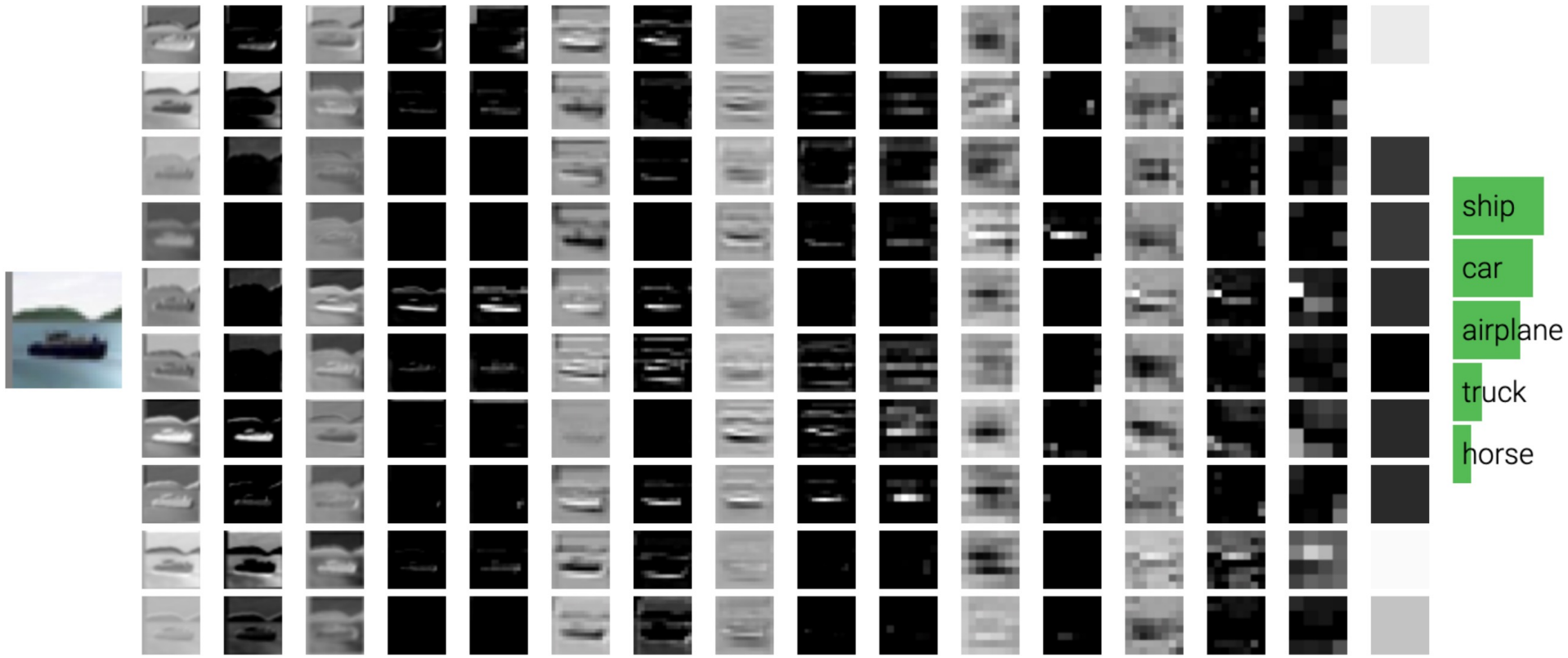
Multi-classification

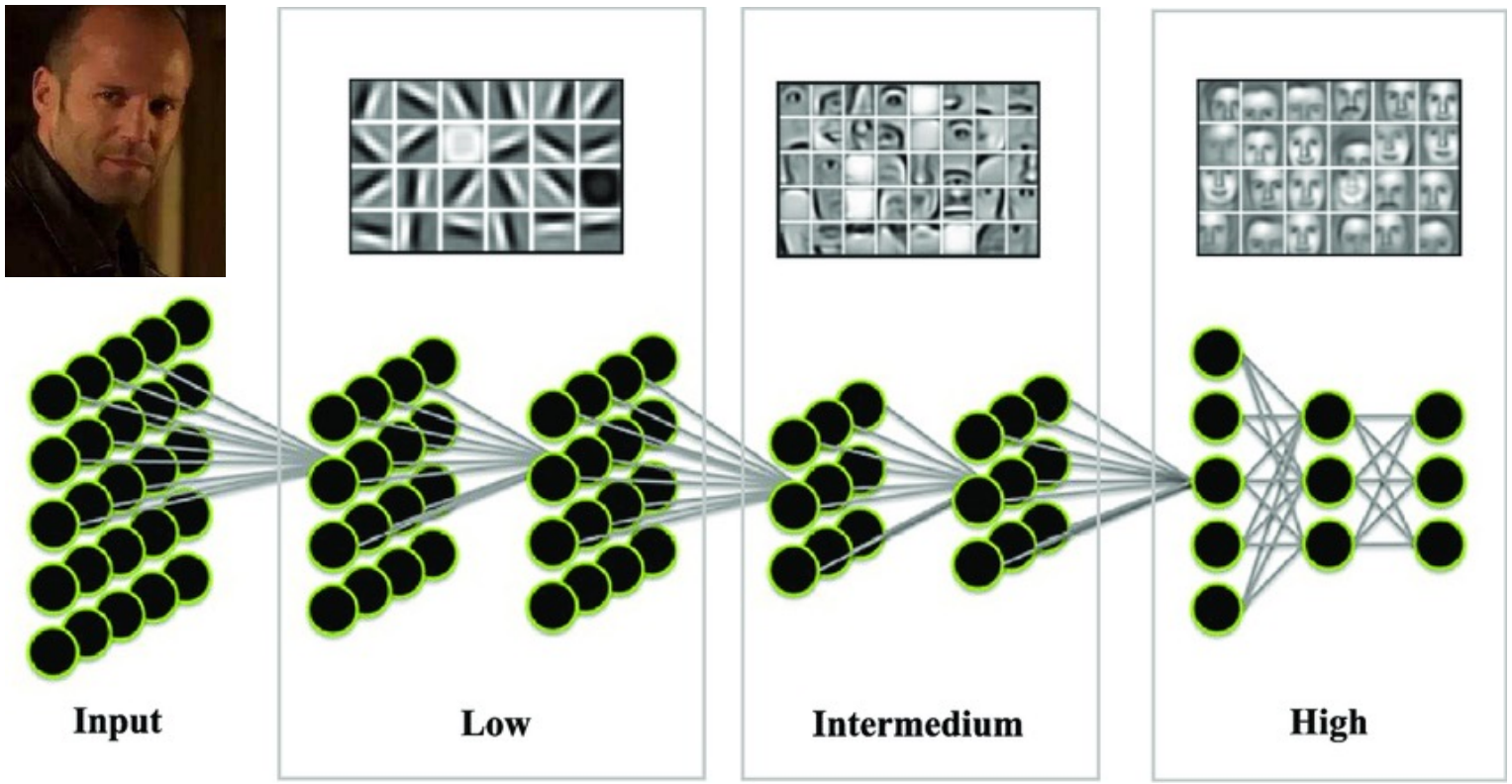
- Multi-classification is very common in real life

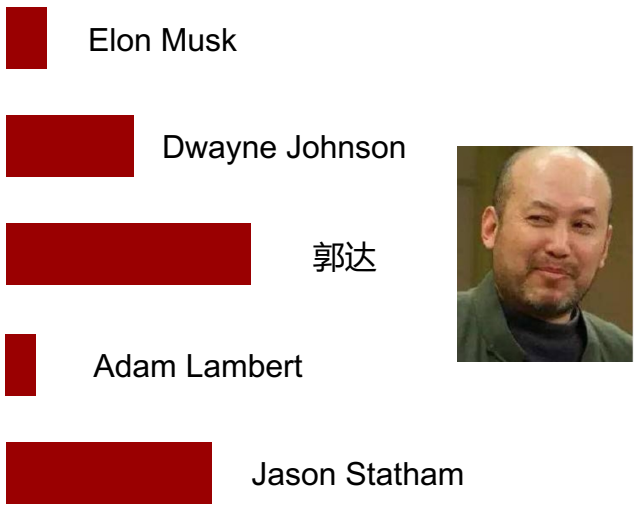


[CS231n: Deep Learning for Computer Vision]

Multi-classification

- Face recognition is one of the most successful multi-classification tasks





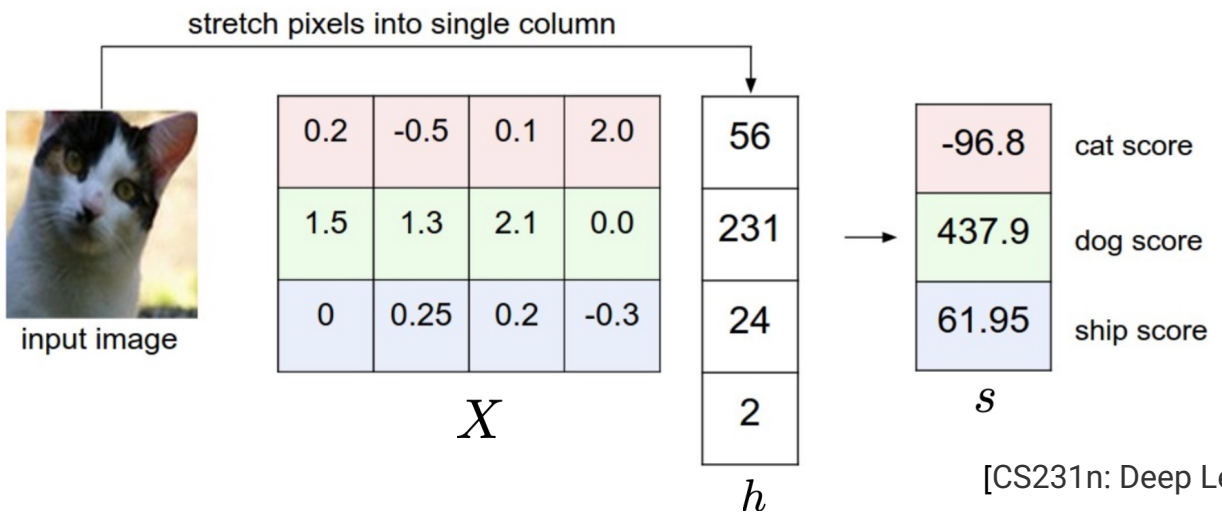
Predict the score

- We first collect the dataset $\{h_i, y_i\}_{i=1}^{N}$ where $h_i \in R^d$ is the feature and $y_i \in \{1, \cdots, C\}$ is the label

- We consider a linear model to predict the score of each class

$$s = Xh \in \mathbb{R}^C$$

where $X \in \mathbb{R}^{C \times d}$ is the model parameters to learn and s_i is the score that h belongs to class i

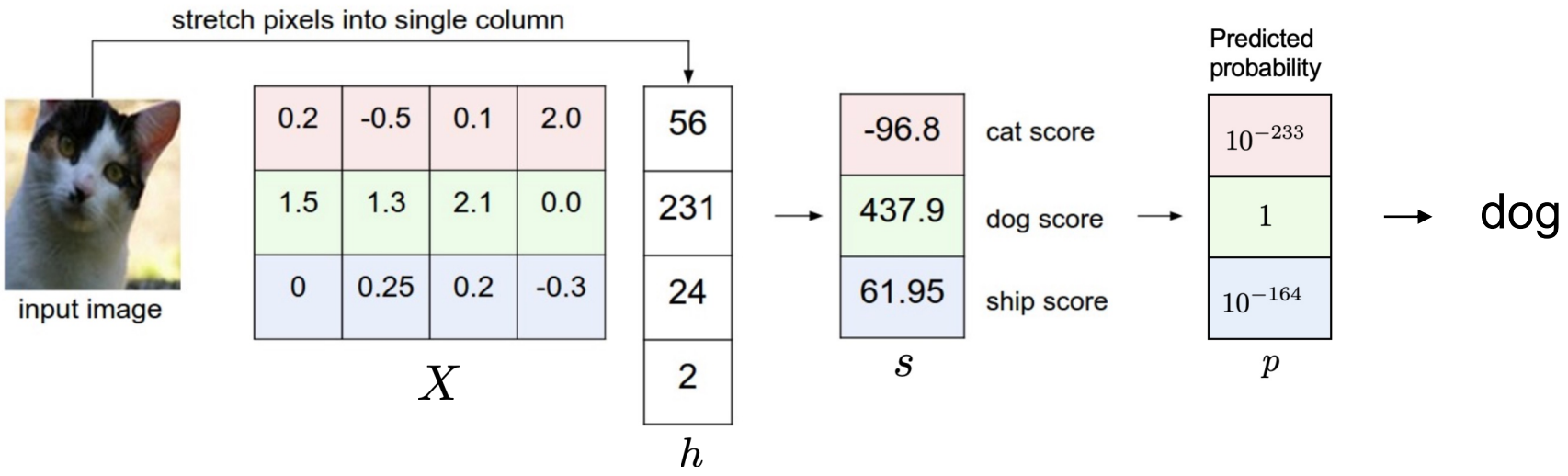


[CS231n: Deep Learning for Computer Vision]

Predict the probability

- Given the score vector s, the probability of each class with the softmax function is as follows

$$p_i = \frac{\exp(s_i)}{\sum_{j=1}^{C} \exp(s_j)} \in (0, 1)$$



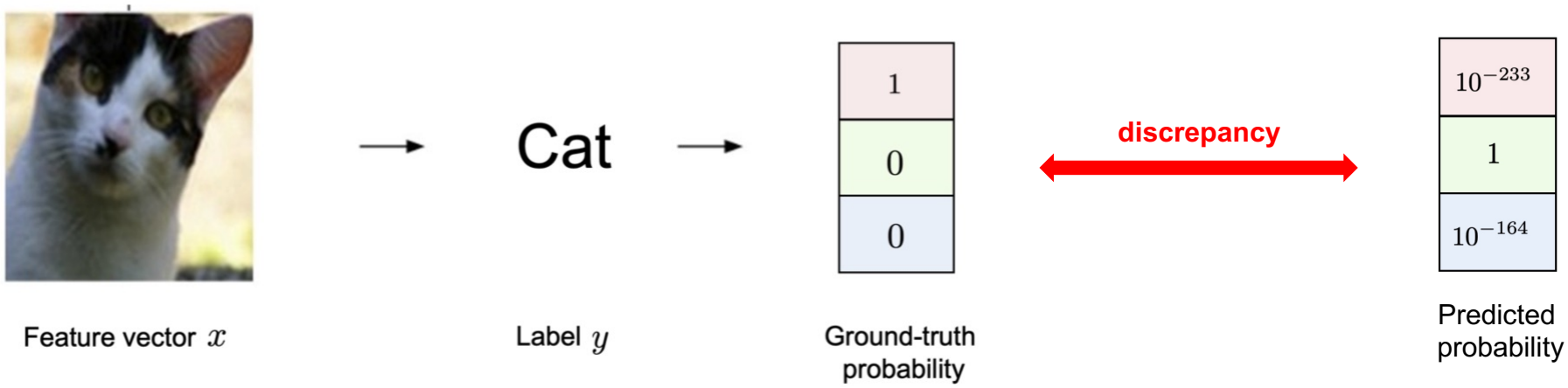
- Given the model parameters X, we can predict the class that feature h belongs to

- But a bad model X will result in incorrect predictions

Cross-entropy

- How to achieve a good model X that can provide accurate predictions?

- We need to train the model so that the discrepancy between ground-truth and predictions are minimized



- How to measure the discrepancy?

Cross-entropy

- Cross entropy can measure the difference between two distributions $p \in R^d$ and $q \in R^d$

$$H(\boldsymbol{p}, \boldsymbol{q}) = -\sum_{j=1}^{d} p_j \log(q_j)$$

Smaller cross entropy indicates smaller difference between p and q

- Examples:

$$\boldsymbol{p} = (1, 0, 0, 0) \quad \boldsymbol{q} = (0.25, 0.25, 0.25, 0.25) \quad \longrightarrow \quad H(\boldsymbol{p}, \boldsymbol{q}) = 2$$

$$\boldsymbol{p} = (1, 0, 0, 0) \quad \boldsymbol{q} = (0.91, 0.03, 0.03, 0.03) \quad \longrightarrow \quad H(\boldsymbol{p}, \boldsymbol{q}) = 0.136$$

Multi-classification: Loss function

- A good model X will minimize the discrepancy between predictions and ground-truth

$$X^\star = \arg\min_X \left\{ \frac{1}{N} \sum_{i=1}^{N} H(p_i, q_i) \right\}, \quad \text{where} \quad p_i = \text{softmax}(X h_i)$$

prediction **ground-truth** **feature of the i-th sample**

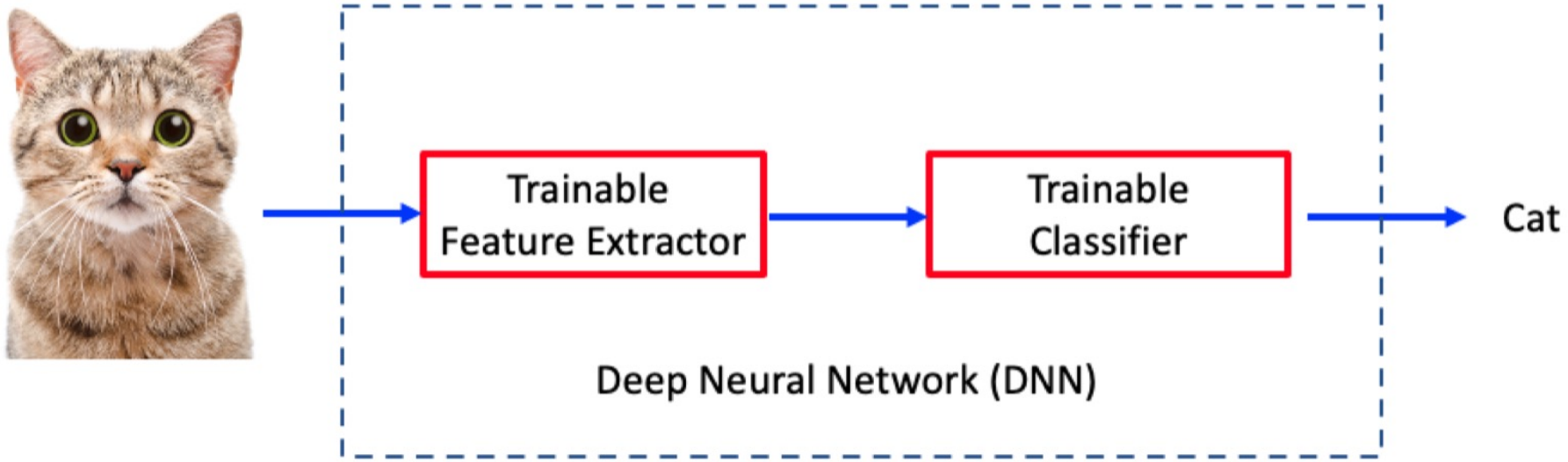
- Once a good model $X^\star$ is achieved, we can make predictions as follows

$$p = \text{softmax}(X^\star h) \longrightarrow h \text{ belongs to class } j^\star = \arg\max_j \{p_j\}$$

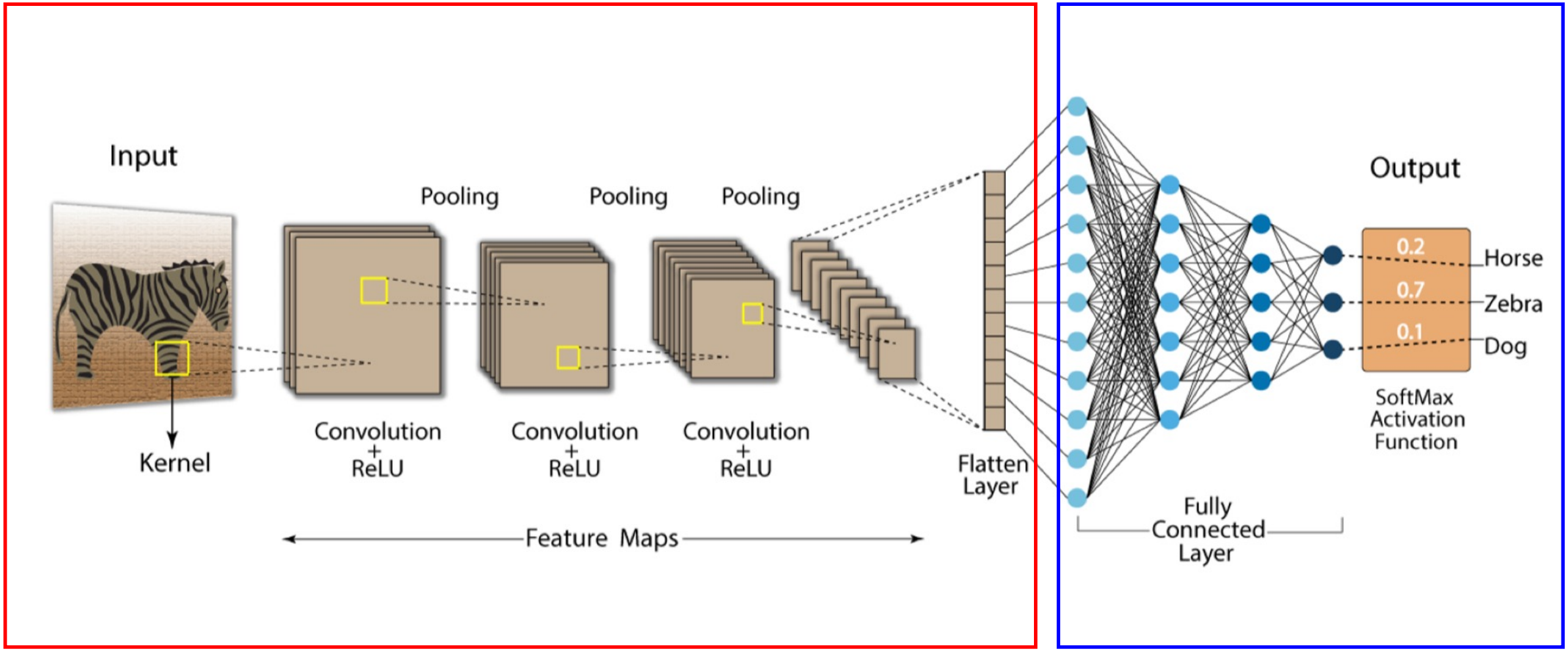
Deep neural network (DNN)

- A deep neural network (DNN) typically includes a **feature extractor** and a **classifier**

- Well-trained DNN can make precise predictions



Example: convolutional neural network

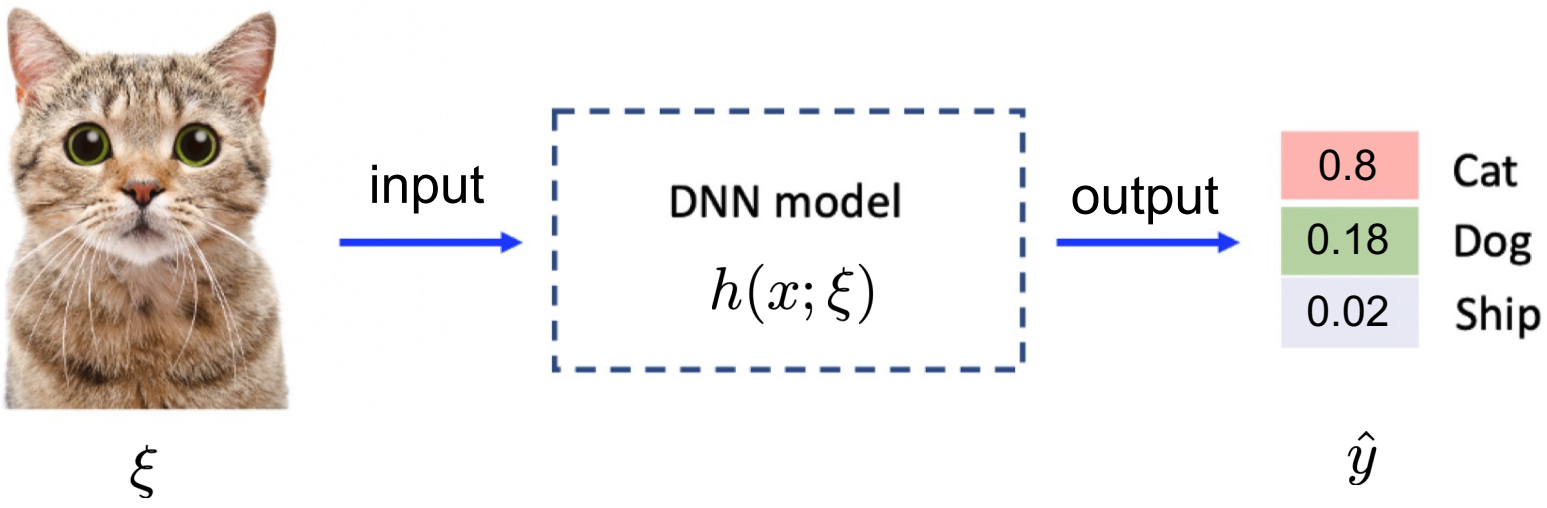


Feature extractor

Multi-Classifier

DNN model

- We model DNN as $h(x;\xi) : \mathbb{R}^d \to \mathbb{R}^c$ that maps input data ξ to a probability $\hat{y}$
  - $x \in \mathbb{R}^d$ is the DNN model parameter to be trained
  - ξ is a random input data sample
  - c is the number of classes



Training DNN can be formulated into an optimization problem

- Define $L(\hat{y}, y) = -\sum_{j=1}^{d} y_{[j]} \log(\hat{y}_{[j]})$ as the loss function to measure the difference between predictions and the ground-truth label, where $y_{[j]}$ is the j-th element in y

- The model parameter $x^\star$ can be achieved by solving the following optimization problem

$$x^\star = \arg\min_{x \in \mathbb{R}^d} \left\{ \mathbb{E}_{(\xi, y) \sim \mathcal{D}} \Big[L\big(h(x; \xi), y\big) \Big] \right\}$$

data distribution **prediction** **real label**

Training DNN can be formulated into an optimization problem

- The model parameter $x^\star$ can be achieved by solving the following optimization problem

$$x^\star = \arg\min_{x \in \mathbb{R}^d} \left\{ \mathbb{E}_{(\xi,y)\sim\underbrace{\mathcal{D}}_{\text{data distribution}}} \left[L\left(\underbrace{h(x;\xi)}_{\text{prediction}}, \underbrace{y}_{\text{real label}} \right) \right] \right\}$$

- If we define $\boldsymbol{\xi} = (\xi, y)$ and $F(x;\boldsymbol{\xi}) = L(h(x;\xi), y)$, the above problem becomes

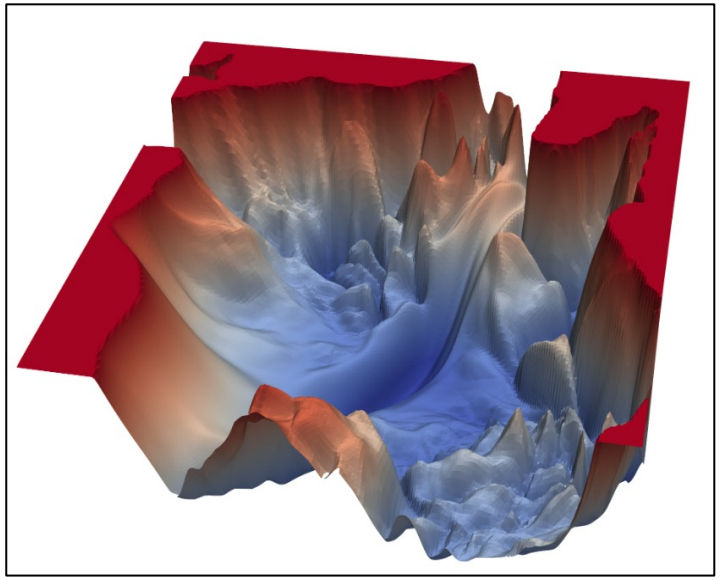
$$x^\star = \arg\min_{x \in \mathbb{R}^d} \left\{ \mathbb{E}_{\boldsymbol{\xi}\sim\mathcal{D}} [F(x;\boldsymbol{\xi})] \right\}$$

- Most optimization researchers use the second formulation as the staring point to develop algorithms

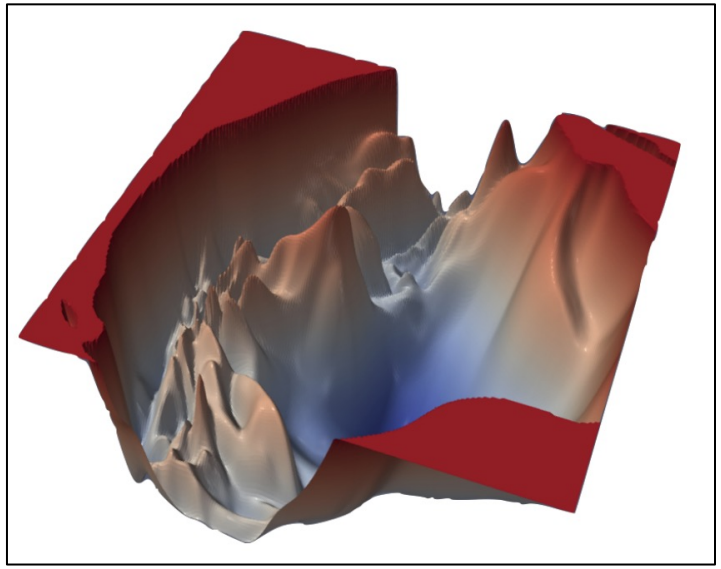
DNN model is notoriously difficult to train

- Highly-nonconvex cost functions; cannot find global minima; trapped into local minimum

VGG-56



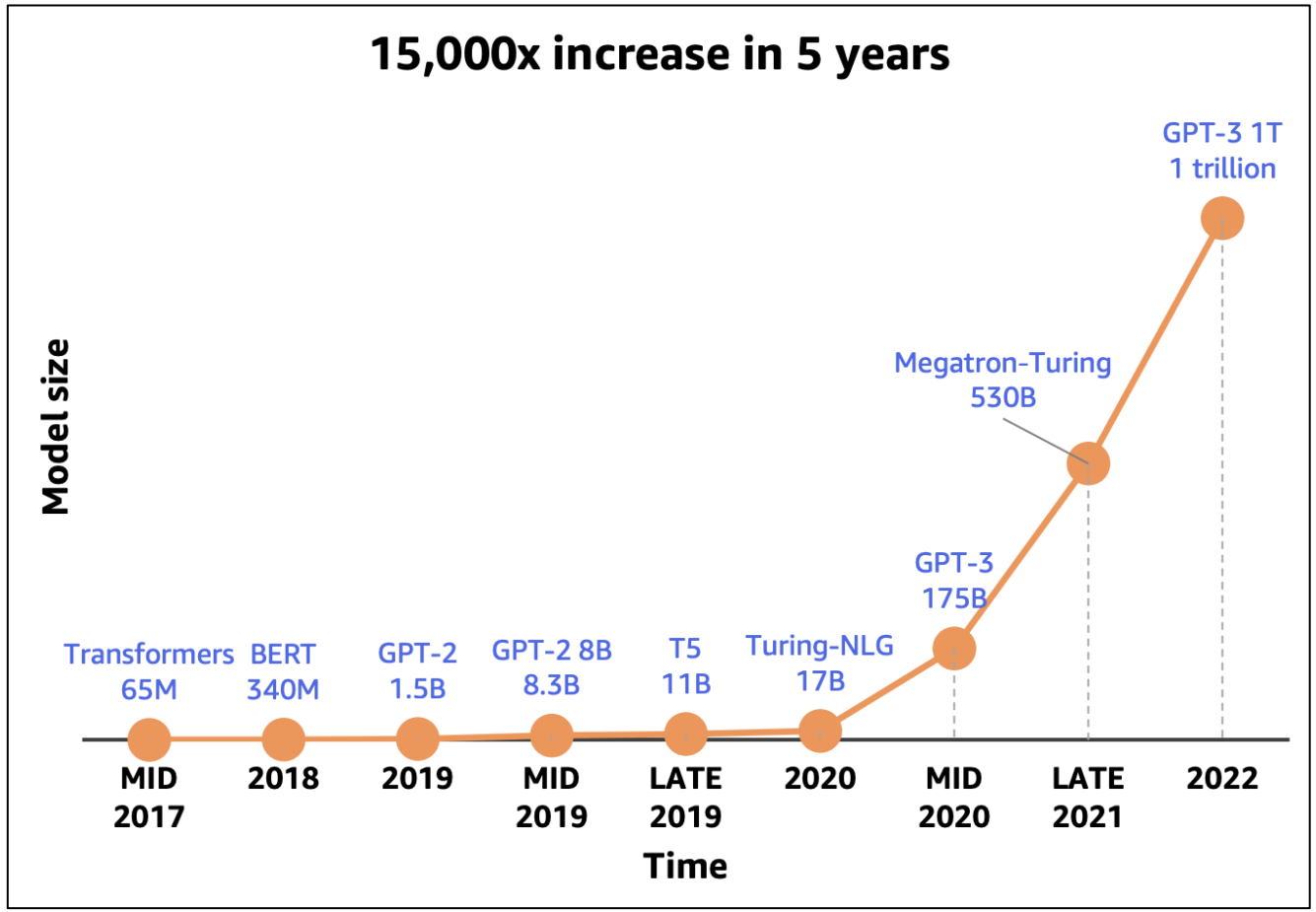
VGG-110



[Visualizing the loss landscape of neural nets]

DNN model is notoriously difficult to train

- The model size is large, i.e., $x \in \mathbb{R}^d$ is of extremely high dimensions



[Train and deploy large language models on Amazon SageMaker]

DNN model is notoriously difficult to train

- The size of the dataset is huge

Dataset	Sampling prop.	Epochs	Disk size
CommonCrawl	67.0%	1.10	3.3 TB
C4	15.0%	1.06	783 GB
Github	4.5%	0.64	328 GB
Wikipedia	4.5%	2.45	83 GB
Books	4.5%	2.23	85 GB
ArXiv	2.5%	1.06	92 GB
StackExchange	2.0%	1.03	78 GB

Table 1: **Pre-training data.** Data mixtures used for pre-training, for each subset we list the sampling proportion, number of epochs performed on the subset when training on 1.4T tokens, and disk size. The pre-training runs on 1T tokens have the same sampling proportion.

Crawled data from websites; in both high quality and low quality

High-quality data

[LLaMA: Open and Efficient Foundation Language Models]

DNN model is notoriously difficult to train

DNN training = Non-convex training + Huge dimensions + Huge dataset

Efficient and **scalable** pretraining approaches are in urgent need